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Lab 9.4 Machine Learning 2 - Machine Learning Implementation to Edge Device

Working with MySQL Database and Grafana Dashboard

4.1 MySQL Database

Let's move on to the data aggregation for the MySQL database. Recall Lab6 and Lab7 for this part. The information of the MySQL server and database is below.

- DNS: mepotrb16.ecn.purdue.edu
- Port: 3306
- Account: firstnamelastname (e.g., johndoe)
- Database: ME597Spring26
- Table: firstnamelastname_lab6 (e.g., johndoe_lab6)
    - Reuse the Table you created in lab6.

The data collector Python script (mtconnect_collecor.py) using XML parsing is prepared for you as Lab7. Follow the steps below to collect all data stream from the MTConnect agent.

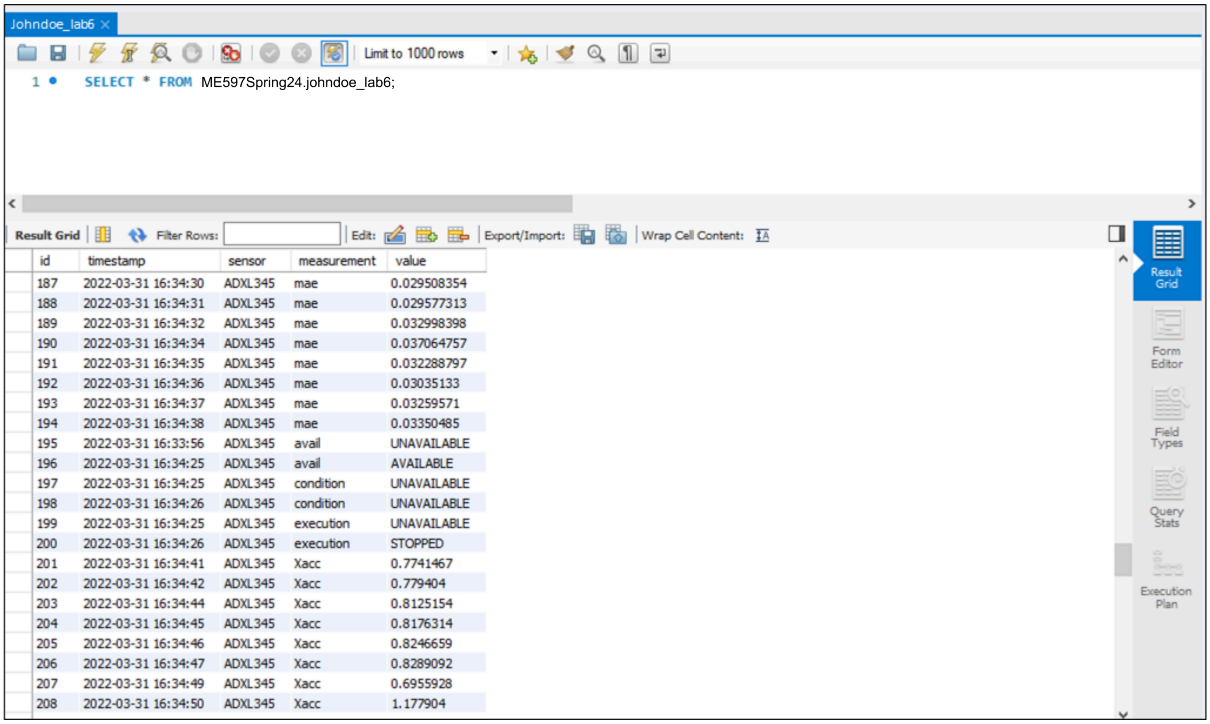
1. Run the sample data collector Python script (mtconnect_collecor.py) on either laptop or Raspberry Pi to store all data stream from the MTConnect agent.

a. You need to modify some front parts that are wrapped in asteriks below. 2. Check if data is being stored using 'MySQL Workbench' on laptop.

Python - Python3 (mtconnect_collecor.py)

```
# Credential
*HOST* = 'MySQL HOST DNS' # MySQL server host DNS
*PORT* = 3306 # MySQL server port number
*USER* = 'your account' # MySQL account name
*PASSWORD* = 'your password' # Password of the account
*DB* = 'Database name' # DB name
*TABLE* = 'your table name' # table name
## Credential

## MTConnect info.
*agent* = "agent host"
*agent_port* = "agent port number"
url_current = "http://"+agent+":"+agent_port+"/current"
MTCONNECT_STR = ET.fromstring(requests.get(url_current).content).tag.split("}")[0]+"}"
print("MTConnect string header is {}.".format(MTCONNECT_STR))
## MTConnect info
```

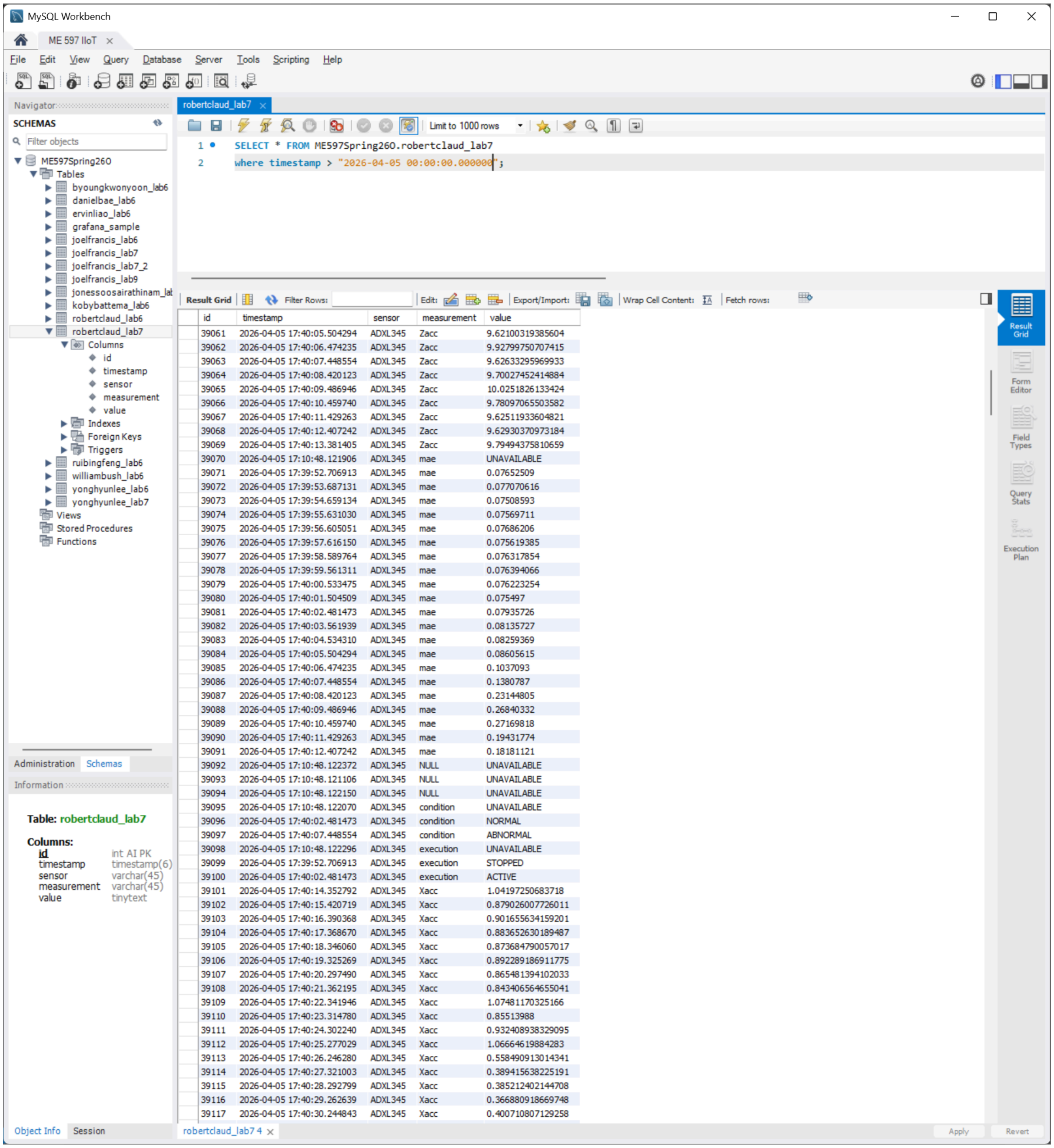


id	timestamp	sensor	measurement	value
187	2022-03-31 16:34:30	ADXL345	mae	0.029508354
188	2022-03-31 16:34:31	ADXL345	mae	0.029577313
189	2022-03-31 16:34:32	ADXL345	mae	0.032998398
190	2022-03-31 16:34:34	ADXL345	mae	0.037064757
191	2022-03-31 16:34:35	ADXL345	mae	0.032288797
192	2022-03-31 16:34:36	ADXL345	mae	0.03035133
193	2022-03-31 16:34:37	ADXL345	mae	0.03259571
194	2022-03-31 16:34:38	ADXL345	mae	0.03350485
195	2022-03-31 16:33:56	ADXL345	avail	UNAVAILABLE
196	2022-03-31 16:34:25	ADXL345	avail	AVAILABLE
197	2022-03-31 16:34:25	ADXL345	condition	UNAVAILABLE
198	2022-03-31 16:34:26	ADXL345	condition	UNAVAILABLE
199	2022-03-31 16:34:25	ADXL345	execution	UNAVAILABLE
200	2022-03-31 16:34:26	ADXL345	execution	STOPPED
201	2022-03-31 16:34:41	ADXL345	Xacc	0.7741467
202	2022-03-31 16:34:42	ADXL345	Xacc	0.779404
203	2022-03-31 16:34:44	ADXL345	Xacc	0.8125154
204	2022-03-31 16:34:45	ADXL345	Xacc	0.8176314
205	2022-03-31 16:34:46	ADXL345	Xacc	0.8246659
206	2022-03-31 16:34:47	ADXL345	Xacc	0.8289092
207	2022-03-31 16:34:49	ADXL345	Xacc	0.6955928
208	2022-03-31 16:34:50	ADXL345	Xacc	1.177904

Figure 12 Result grid of stored data in MySQL workbench

Task 4.1

Capture the result grid as Figure 12 and then attach it to the report.



4.2 Grafana dashboard

Finally, let's create a web-based dashboard using Grafana. Recall Lab8 activities. The information of the Grafana server domain and the port number is below. Log in the Grafana server in a web-browser of laptop.

- DNS: mepotrb16.ecn.purdue.edu
- Port: 3000
- Grafana web page URL: http://mepotrb16.ecn.purdue.edu:3000/
- Account (username): yourname (e.g., johndoe)

Go to '*Lab9 sample*' dashboard in '*0_SAMPLE*' dashboard folder. This dashboard (Figure 13) is what you should finally create.

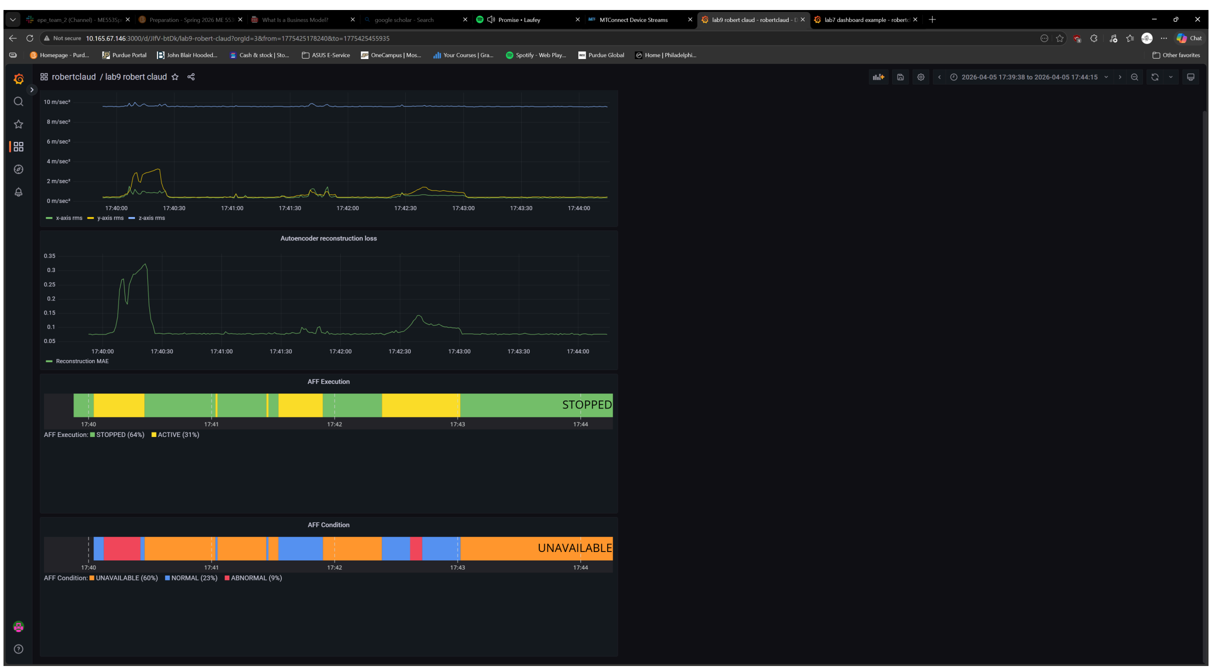
Figure 13 Sample Grafana dashboard (0_SAMPLE/Lab9 sample)

Task 4.2

Create a dashboard as Figure 13 in your folder created in Lab 7 and save it as the dashboard name of 'lab9 your name', e.g., 'lab9 john doe'.

Capture the dashboard and attach it to the report.

After creating the dashboard, confirm if the entire monitoring system works in real-time by refreshing the dashboard.



Lab9 Summary and Deliverables

Answer the following questions for your achievements

Q1. Please summarize Lab9.

In Lab 9 we deployed a previously trained autoencoder model to an edge device. In this case, the edge device was the Raspberry Pi with a connected ADXL345 accelerometer. From there, we developed an algorithm that could detect an rms threshold to determine if an axial flow fan was turned on or off. Using this threshold, once the AFF was determined to be turned on the autoencoder machine learning model was used to determine if the fan had a normal or abnormal operating condition. Using these conditions as well as the result of the ML model's reconstruction we dumped these values into an SQL database using the MTConnect Agent and Adapter and built a Grafana dashboard to monitor the results.

Q2. What skills did you have to develop to accomplish this project?

To accomplish this project, I needed to develop more of my ML skills and data processing.

Q3. What aspects of this project were the most beneficial for your learning?

The most beneficial aspects of this project were the fact that it was a real-world example that I could touch and interact with. It felt more real than a practice dataset to use machine learning on.

Q4. What challenges did you encounter in completing the project?

The challenge that I encountered was that the max, min and mae threshold values were wrong since the overall operating conditions from lab 8 was different from this lab.

Q5. How did you overcome the challenges or remedy the problems encountered?

To overcome this challenge, I quickly retrained a model under my current operating conditions and used new data to get the correct max, min and mae threshold values.

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